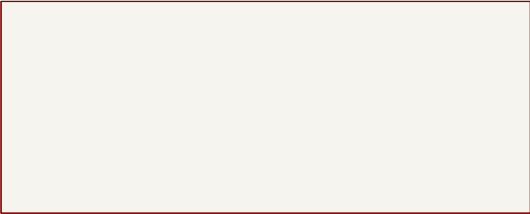


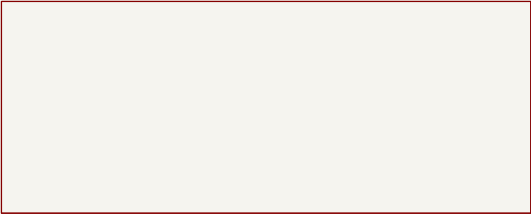
CONTROLLER – SEVEN READABLE KICAD SHEETS

01 Power and Pi



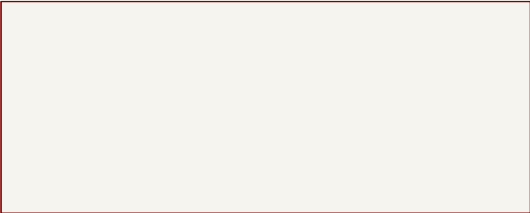
File: 01_power_and_pi.kicad_sch

02 ESP32 and Pi UART



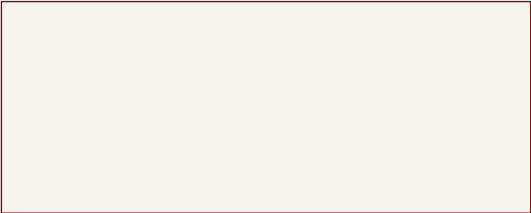
File: 02_esp32_and_pi.kicad_sch

03 Left Motor



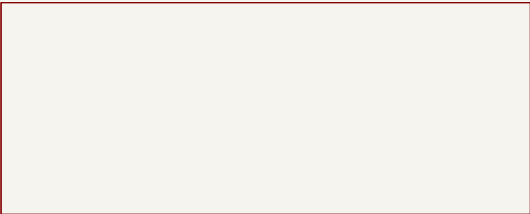
File: 03_left_motor.kicad_sch

04 Right Motor



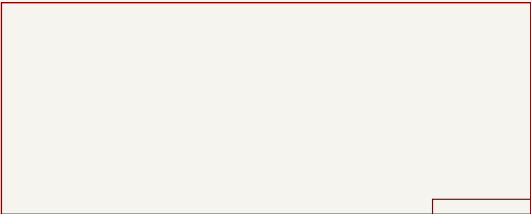
File: 04_right_motor.kicad_sch

05 Encoders



File: 05_encoder_inputs.kicad_sch

06 Arm Servos



File: 06_arm_servos.kicad_sch

07 Main Servo 8.4V



Harrison

Sheet: /
File: pickleball_robot_controller.kicad_sch

Title: Pickleball Robot Controller – Hierarchical Overview

Size: A4	Date: 2026-07-12	Rev: R2-MULTISHEET
KiCad E.D.A. 10.0.4		Id: 1/8

ERY INPUT AND PROTECTION

RASPBERRY PI 5 V / 5 A BUCK

external latching E-stop between F1 and the VBAT_SW distribution point.

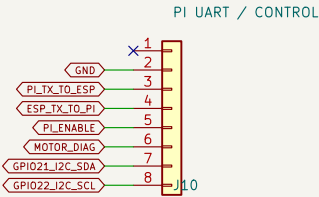
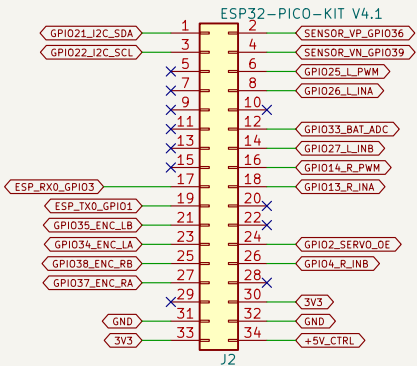
BATTERY VOLTAGE MONITOR

TROL BUCK – ESP32 AND LOGIC ONLY

PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG

CARRIER – CONFIRMED 17.78 mm ROW SPACING

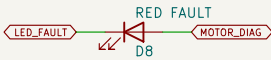
RASPBERRY PI UART / CONTROL – NORMAL 2.54 mm HEADER



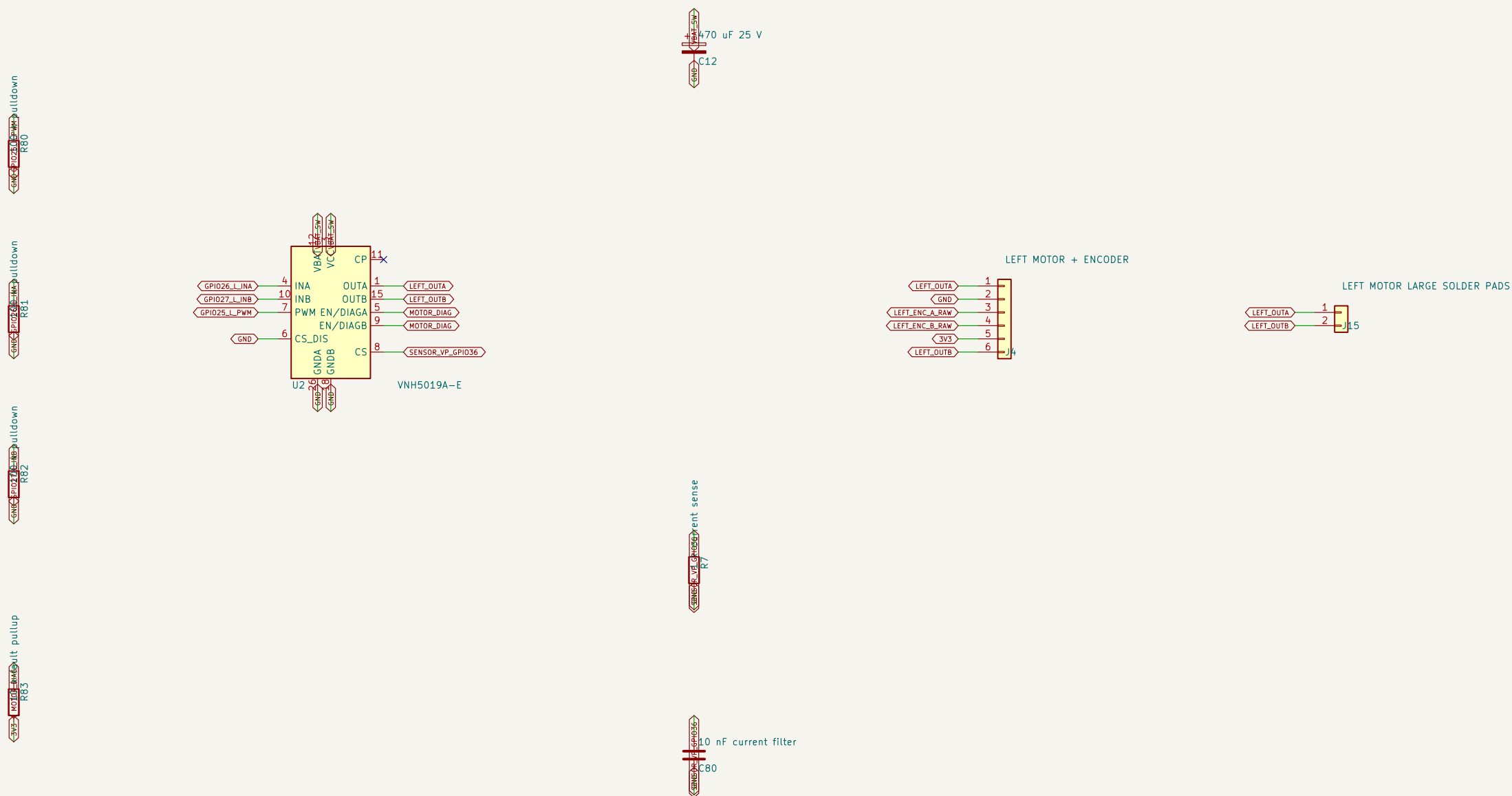
J10 pin 1 UNUSED; pin 2 GND; pin 3 Pi TX; pin 4 Pi RX; pins 5-8 optional control/I2C.

Pi GPIO14 physical pin 8 → J10.3; Pi GPIO15 physical pin 10 ← J10.4; Pi GND → J10.2.

STATUS LEDS



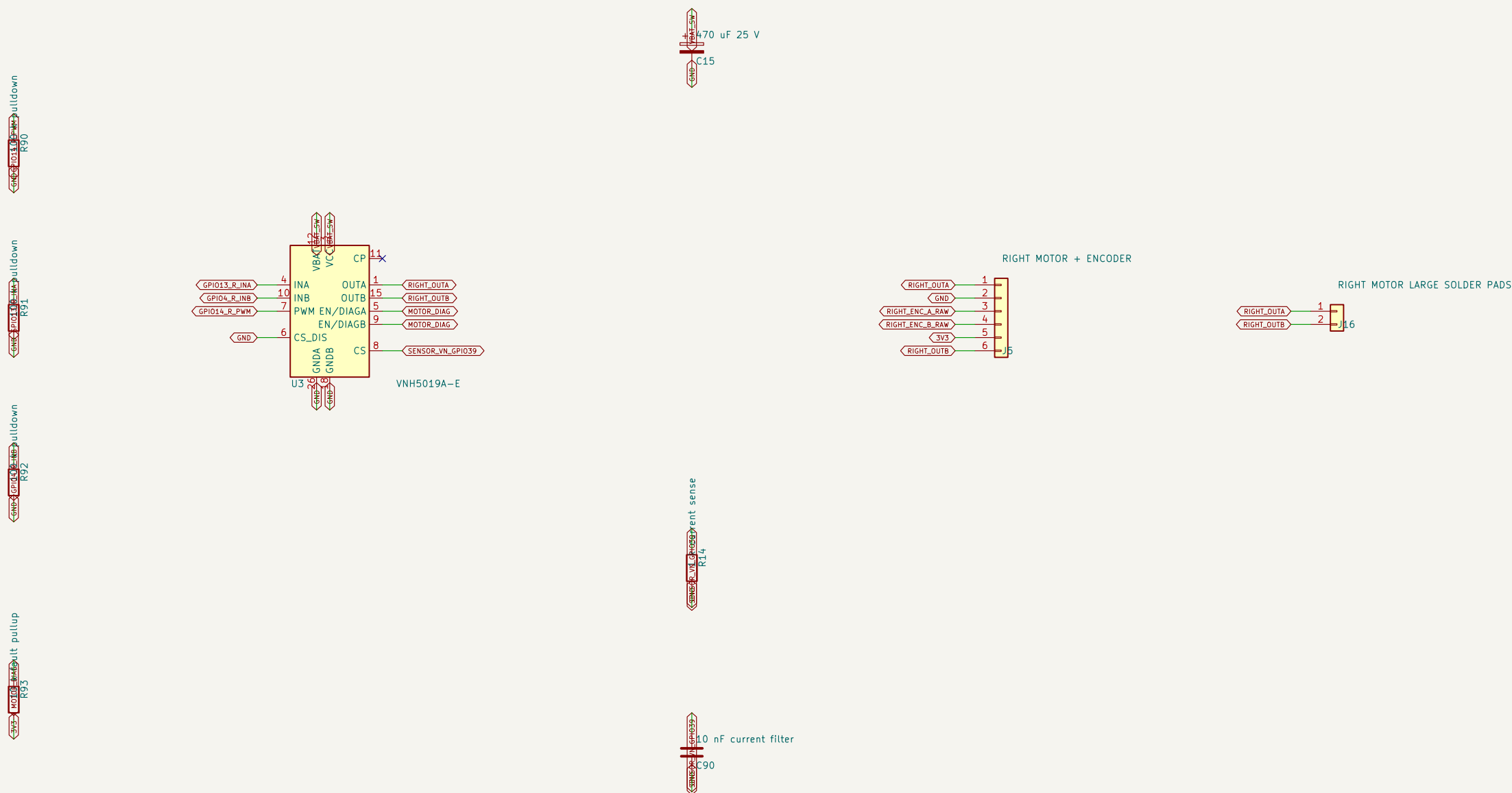
LEFT WHEEL MOTOR DRIVER



< encoder GND; 3 YELLOW encoder A; 4 GREEN encoder B; 5 BLUE encoder +3V3; 6 WHITE motor-.

holes in parallel with pins 1/6 for separately soldered red and white motor-power wires.

RIGHT WHEEL MOTOR DRIVER



< encoder GND; 3 YELLOW encoder A; 4 GREEN encoder B; 5 BLUE encoder +3V3; 6 WHITE motor-.

holes in parallel with pins 1/6 for separately soldered red and white motor-power wires.

SCHMITT-BUFFERED ENCODER INPUTS

A

B

C

D

E

F

A

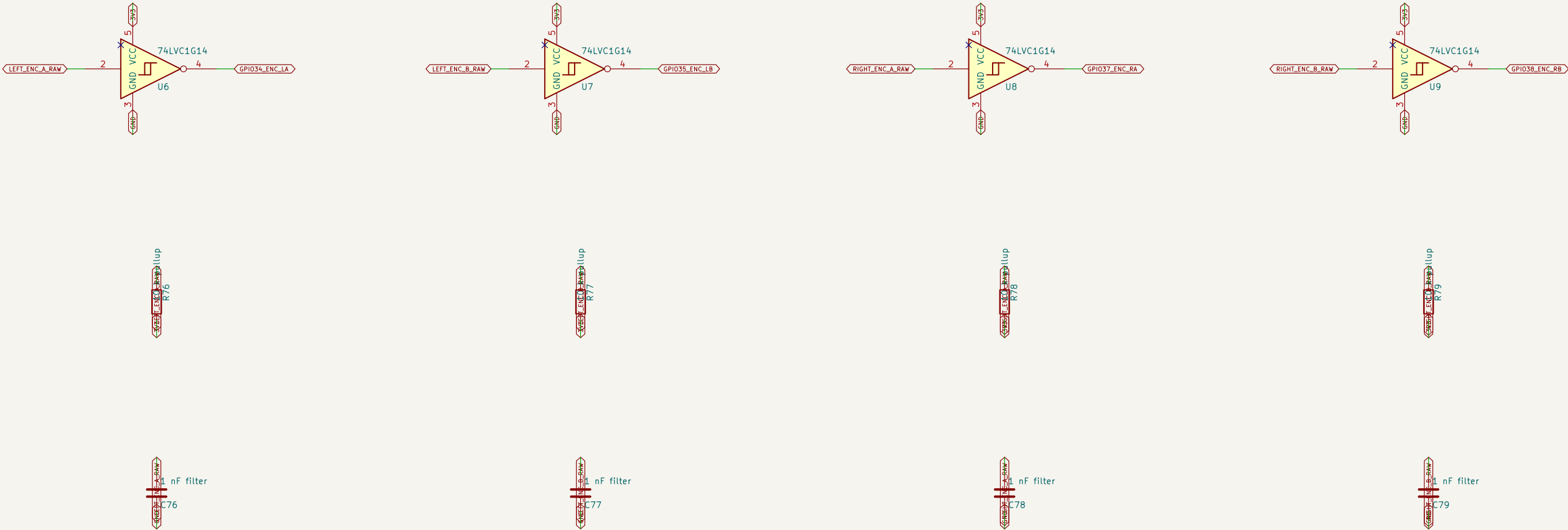
B

C

D

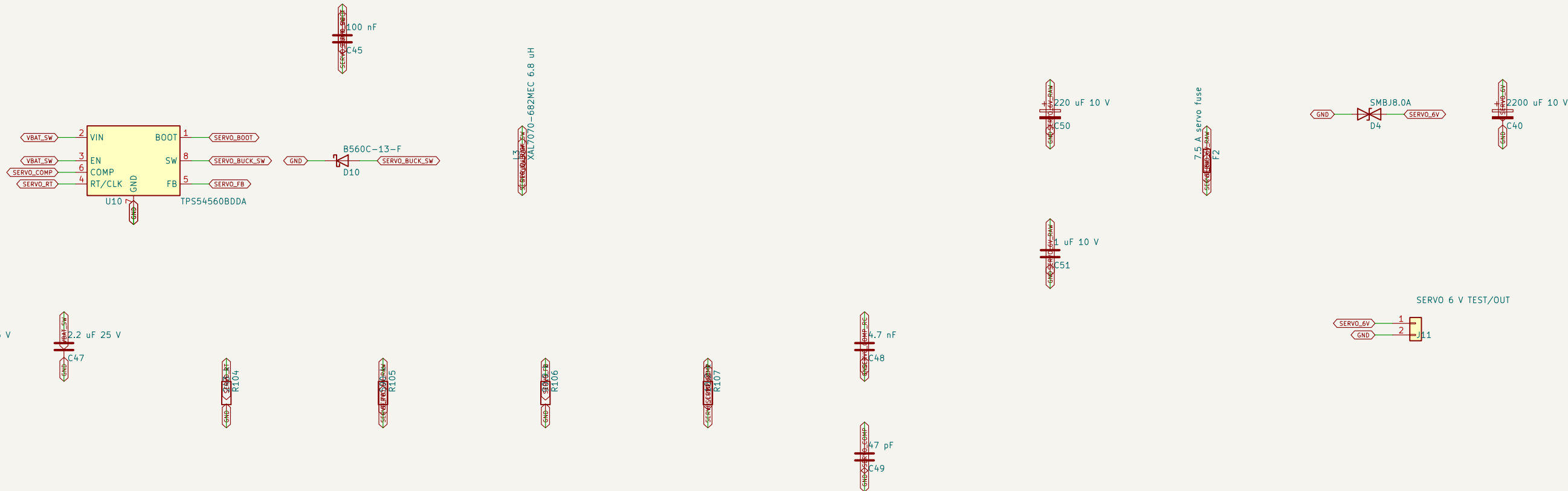
E

F

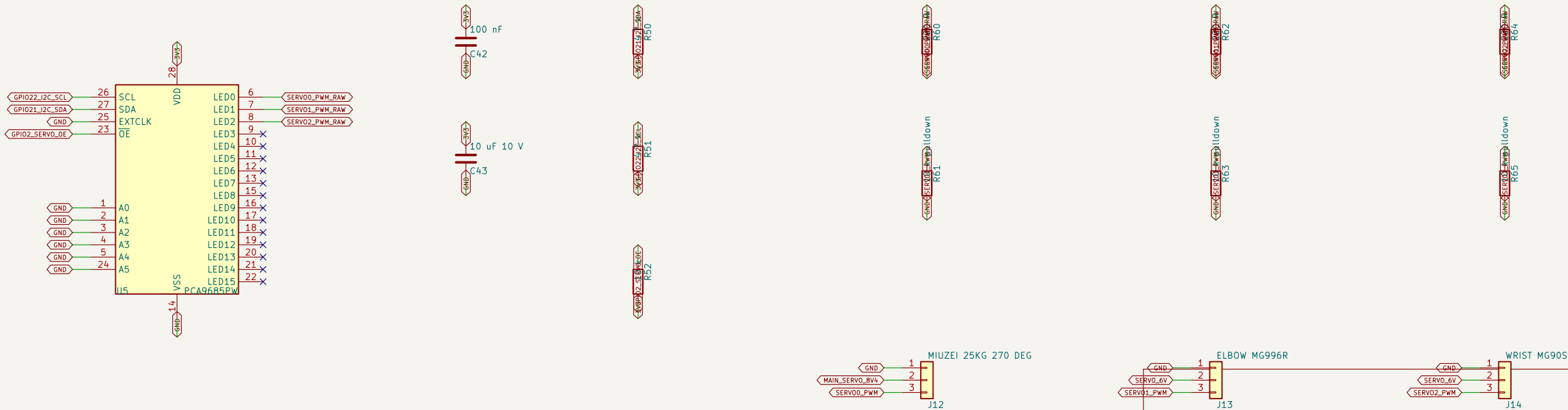


tors are on the motor-driver sheets; their encoder A/B pins enter here.

6 V / 5 A SERVO BUCK



A9685 SERVO PWM CONTROL



PWR_FLAG

Header order: pin 1 GND, pin 2 power, pin 3 PWM, J12 = 8.4 V; J13/J14 = 6 V.

Sheet 6 of 7		
Harrison		
Sheet: /06 Arm Servos/ File: 06_arm_servos.kicad_sch		
Title: Servo Power Regulator and 3-DOF Arm Control		
Size: A3	Date: 2026-07-12	Rev: R2-MULTISHEET
KiCad E.D.A. 10.0.4		Id: 7/8

5 A BUCK FOR MIUZEI 25KG 270 DEG SERVO

ck resistors for approximately 8.42 V. J12 is the only servo powered from this rail.

R and J14 MG90S remain on the separate regulated 6 V rail.